

Glomerular Disease — Nephrotic PEAL & Podocyte Diseases (Podocyte Elementary)



1. PEAL mnemonic

WHAT YOU SEE

Chalkboard — P E A L list

WHAT IT ENCODES

PEAL captures the nephrotic tetrad: Proteinuria >3.5 g/day, Edema, hypoAlbuminemia (<3 g/dL), and hyperLipidemia. The unifying mechanism is loss of glomerular barrier charge/size selectivity, so proteinuria is the primary event and the other three follow from low oncotic pressure. Add the complications layer — hypercoagulability (antithrombin III loss) and infection risk (Ig loss) — for the full picture.

BOARD TRAP

WRONG answer: adding hematuria or RBC casts to the nephrotic syndrome definition. Those are NEPHRITIC features (glomerular inflammation/bleeding); the discriminator is that PEAL is a barrier-permeability problem with a bland sediment, not an inflammatory one.

2. Proteinuria >3.5 g/day

WHAT YOU SEE

P = overflowing funnel >3.5 g/day

WHAT IT ENCODES

Nephrotic-range proteinuria is >3.5 g/24h, reflecting failure of the size/charge barrier (GBM heparan sulfate + podocyte slit diaphragm). It is quantified by 24h collection or spot urine protein:creatinine ratio (>3-3.5 g/g). 'Selective' proteinuria (mostly albumin) suggests MCD, whereas 'non-selective' loss of larger globulins suggests more structural barrier damage (FSGS/membranous).

BOARD TRAP

WRONG answer: classifying 1-3 g/day as nephrotic. That is sub-nephrotic; nephritic syndrome typically runs <3.5 g/day WITH hematuria and RBC casts — the discriminator is magnitude of protein loss plus sediment activity.

3. Albumin lost

WHAT YOU SEE

Yellow puddle — albumin lost on factory floor

WHAT IT ENCODES

Massive transglomerular albumin escape exceeds hepatic compensation, producing hypoalbuminemia (<3 g/dL) and the fall in oncotic pressure that drives edema and reactive hyperlipidemia. Loss of albumin also reduces carrier capacity for protein-bound drugs, hormones, and vitamin D-binding protein. In MCD the loss is highly selective for albumin, reflecting predominantly a charge-barrier defect.

BOARD TRAP

WRONG answer: assuming all nephrotic albumin loss is equally selective. Highly SELECTIVE albuminuria points specifically to MCD; non-selective loss of larger proteins suggests FSGS or membranous — the discriminator is the protein-selectivity profile on the biopsy/urine.

4. Thrombosis — antithrombin III loss

WHAT YOU SEE

Red clot machine + RVT / post-cava

WHAT IT ENCODES

Urinary loss of antithrombin III, along with increased hepatic procoagulants (fibrinogen) and platelet hyperactivity, creates a hypercoagulable state. The signature complication is renal vein thrombosis — flank pain, hematuria, sudden worsening proteinuria, and possible left varicocele — most associated with membranous nephropathy; DVT/PE also occur. Anticoagulate when albumin is very low (<2-2.5 g/dL) or risk is high.

BOARD TRAP

WRONG answer: blaming thrombocytosis/platelet excess. The driver is ANTITHROMBIN III loss in the urine (a loss-of-anticoagulant state), not a primary platelet disorder — the discriminator is low antithrombin activity and the membranous association for RVT.

5. Dyslipidemia

WHAT YOU SEE

Chalkboard — dyslipidemia, increased LDL

WHAT IT ENCODES

Reduced plasma oncotic pressure upregulates hepatic lipoprotein synthesis and impairs lipid clearance (reduced LCAT and LDL-receptor activity), raising LDL, total cholesterol, and triglycerides. The resulting atherogenic profile increases cardiovascular risk and contributes to lipiduria/fatty casts and Maltese-cross bodies on polarized microscopy. Treat with a statin (e.g., atorvastatin 10-40 mg daily).

BOARD TRAP

WRONG answer: predicting low cholesterol/triglycerides. Lipids RISE in nephrotic syndrome via hepatic overproduction — the discriminator from a starvation/malabsorption picture is the coexisting proteinuria, edema, and hypoalbuminemia.

6. Infection / altered drug binding

WHAT YOU SEE

Chalkboard — infection, altered drug t1/2

WHAT IT ENCODES

Urinary loss of immunoglobulins (IgG) and complement factor B raises susceptibility to ENCAPSULATED organisms — classic pneumococcal spontaneous bacterial peritonitis in a nephrotic child with ascites. Concurrent loss of albumin and binding globulins alters pharmacokinetics: the free fraction of highly protein-bound drugs rises and dosing/half-life shift, and vitamin D-binding protein and thyroxine-binding globulin losses can cause functional deficiencies.

BOARD TRAP

WRONG answer: treating altered protein binding as a clinically trivial footnote, or assuming infection risk reflects a separate immunodeficiency. The mechanism is acquired urinary protein loss — the discriminator is hypoalbuminemia altering free drug levels and Ig loss driving encapsulated-organism infection.

7. MCD — 70-90% pediatric nephrotic

WHAT YOU SEE

Podocyte Elementary — Nil lesion classroom

WHAT IT ENCODES

Minimal change ('nil') disease causes 70-90% of pediatric nephrotic syndrome: normal light microscopy, negative immunofluorescence, and diffuse foot-process effacement only on EM. It is highly steroid-responsive — children are usually treated empirically with prednisone (e.g., 60 mg/m²/day) without biopsy. Triggers include recent viral URI, atopy, NSAIDs, and Hodgkin lymphoma.

BOARD TRAP

WRONG answer: treating MCD as a common ADULT cause. In adults it is only ~10-15% of nephrotic syndrome; FSGS leads in adults — the discriminator is the pediatric predominance and brisk steroid responsiveness of MCD.

8. Foot-process effacement

WHAT YOU SEE

EM podocyte feet flattened

WHAT IT ENCODES

Effacement (flattening/fusion) of podocyte foot processes seen only on electron microscopy is the unifying ultrastructural lesion of MCD and a feature of FSGS. The podocyte slit diaphragm proteins — nephrin and podocin — normally form the final size barrier; their disruption causes the foot processes to retract and the barrier to leak protein. In pure MCD the light microscopy and IF are normal, so EM is required.

BOARD TRAP

WRONG answer: attributing foot-process effacement to subepithelial spikes/deposits. Subepithelial 'spike-and-dome' deposits with granular IgG/C3 define MEMBRANOUS nephropathy; foot-process effacement without deposits points to MCD/FSGS — the discriminator is the presence or absence of immune deposits on EM/IF.

9. Steroids first-line (MCD)

WHAT YOU SEE

Prednisone bottle in classroom

WHAT IT ENCODES

In children, MCD is treated empirically with corticosteroids (prednisone ~60 mg/m²/day, tapered over weeks) and biopsy is typically deferred because steroid response is the norm; relapses may be managed with calcineurin inhibitors, cyclophosphamide, or rituximab. The brisk response reflects the reversible, immune-mediated podocyte injury without structural scarring.

BOARD TRAP

WRONG answer: applying empiric-steroids-without-biopsy to adults. That approach is appropriate in children with classic MCD, but adults with nephrotic syndrome generally need a biopsy first because FSGS, membranous, and secondary causes differ in treatment — the discriminator is patient age and atypical features.

10. FSGS variants

WHAT YOU SEE

Basement lineup — NOS, tip, cellular, perihilar, collapsing

WHAT IT ENCODES

FSGS histologic variants carry distinct prognoses: NOS (classic, default), tip lesion (BEST prognosis, most steroid-responsive), cellular, perihilar (linked to obesity/hyperfiltration and reduced nephron mass), and collapsing (WORST prognosis, associated with HIV and APOL1 risk alleles). All share focal/segmental sclerosis on light microscopy and foot-process effacement on EM, and many are steroid-resistant.

BOARD TRAP

WRONG answer: reversing the prognostic poles. Collapsing variant has the WORST outcome and tip lesion the BEST — flipping them is the classic distractor; the discriminator is variant histology plus the HIV/APOL1 link for collapsing.

11. APOL1 / African ancestry risk

WHAT YOU SEE

APOL1 risk + African ancestry plaque

WHAT IT ENCODES

Two APOL1 high-risk variants (G1/G2), common in people of West African ancestry, markedly increase risk of FSGS — particularly the collapsing variant and HIV-associated nephropathy (HIVAN). The same alleles accelerate progression of nondiabetic CKD to ESKD. They confer ancestral protection against trypanosomiasis, explaining their high frequency.

BOARD TRAP

WRONG answer: linking APOL1 to MCD or membranous nephropathy. APOL1 risk variants are tied specifically to FSGS/collapsing/HIVAN — the discriminator is the FSGS histology and African ancestry, not a podocyte-effacement-only lesion or immune-deposit disease.

12. FSGS — ESKD #1 primary cause

WHAT YOU SEE

Red ESKD #1 sign — ~50% to ESKD in 10 yrs

WHAT IT ENCODES

FSGS is a leading primary glomerular cause of progression to end-stage kidney disease, with roughly half of patients reaching ESKD within about 10 years, especially when nephrotic-range and steroid-resistant. Progression reflects irreversible segmental scarring and nephron loss, contrasting with the generally benign, non-progressive course of MCD. It can also recur in renal allografts, sometimes within hours to days of transplant.

BOARD TRAP

WRONG answer: predicting frequent ESKD progression from MCD. MCD rarely scars or progresses to ESKD; sustained, steroid-resistant nephrotic syndrome that marches toward ESKD is FSGS — the discriminator is segmental sclerosis on biopsy and steroid resistance.